

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I SUBASHREE.N (192521377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWER 1:**Schema**

A **schema** is the overall design or blueprint of a database. It defines how data is organized into tables, columns, relationships, and constraints.

Example:**Student Table**

Student Id	Name	Department
This table structure is the schema		

Instance

An **instance** is the actual data stored in the database at a particular time. It changes whenever data is added, updated, or deleted.

Example:

Students id	Name	Department
101	Ravi	Cse
102	Priya	IT

Types of Schema**1. Logical Schema**

- Describes the logical structure of the database.
- Defines tables, attributes, relationships, and constraints.
- Does not describe storage.

Example:

Student(Student_ID, Name, Department)

2. Physical Schema

- Describes how data is physically stored.
- Includes storage location, indexing, and file organization.
- Used for improving performance.

Example:

Student table stored in File A with B+ Tree Index.

3. View Schema (External Schema)

- Shows only the required data to a particular user.
- Hides unnecessary information.
- Improves security.

Comparison Table

Logical schema	Physics schema	View schema
Defines logical structure	Defines physical storage	Defines user view
Includes tables and relationships	Includes files and indexes	Shows selected data
Used by designers	Used by DBA	Used by end users

ANSWER 2:

Entities and Attributes

1. Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone
- Email

2. Account

- Account_No (Primary Key)
- Account_Type
- Balance
- Opening_Date

3. Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

4. Loan

- Loan_ID (Primary Key)
- Loan_Amount
- Loan_Type

5. Employee

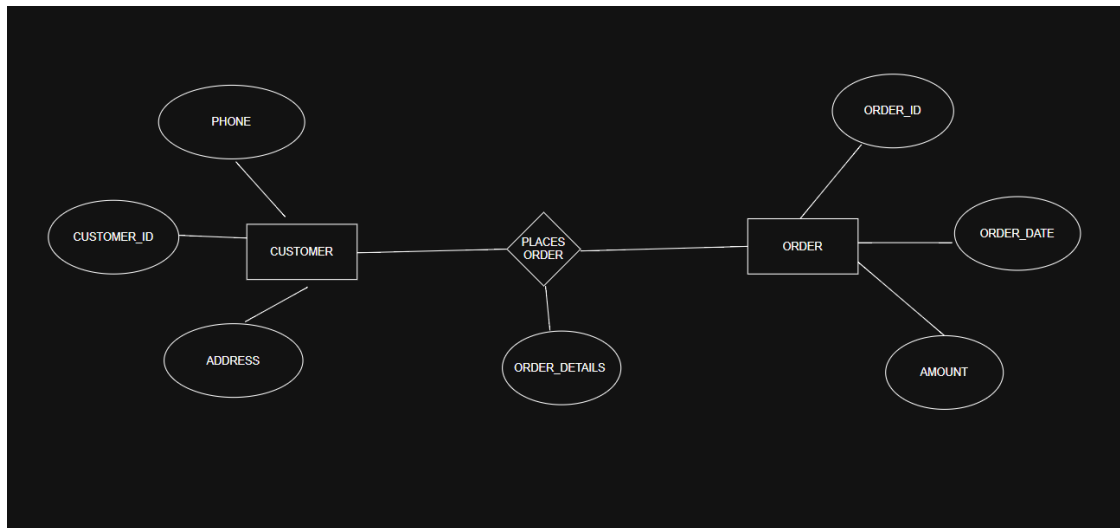
- Employee_ID (Primary Key)
- Name
- Designation
- Salary

Relationships

- Customer **owns** Account (1:M)
- Branch **maintains** Account (1:M)
- Customer **takes** Loan (1:M)

- Branch **issues** Loan (1:M)
- Branch **employs** Employee (1:M)

SIMPLE ER DIAGRAM :



ANSWER 3:

Extended ER (EER) Model

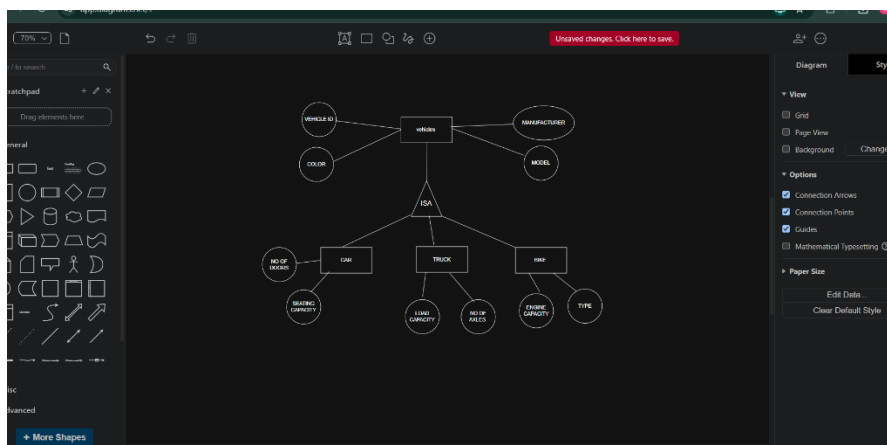
The **Extended Entity Relationship (EER) model** is an advanced version of the ER model. It supports inheritance, subclasses, and more complex relationships.

It is used to represent real-world systems more accurately.

1. Generalization

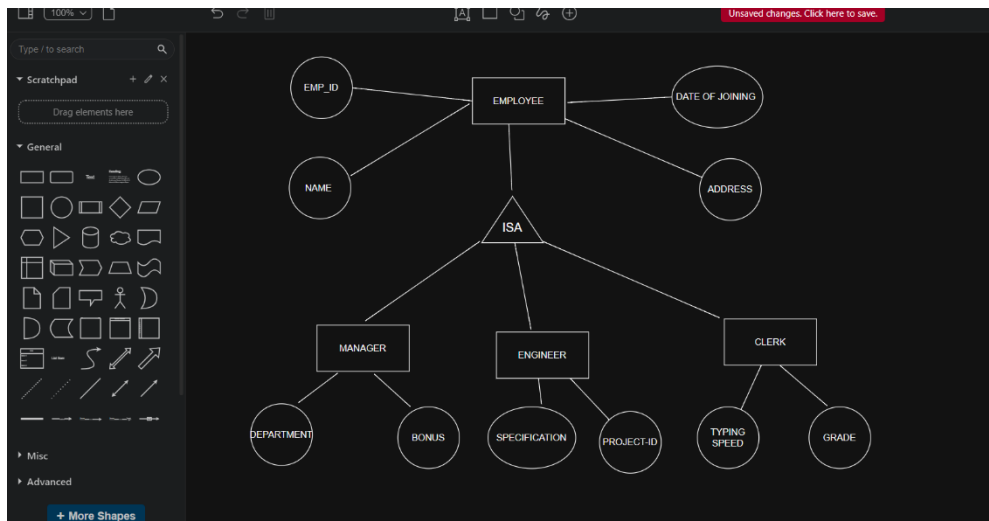
Generalization combines similar lower-level entities into one higher-level entity.

Example:



2. Specialization

Specialization divides one higher-level entity into smaller specialized entities.

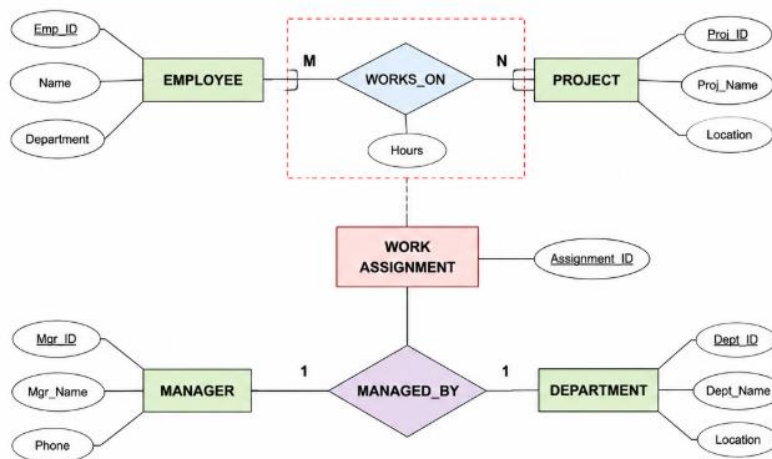


Employee is divided into Manager,engineer and Clerk.

3. Aggregation

Aggregation combines a relationship and an entity into one higher-level entity. It is useful when one relationship participates in another relationship.

Example:



ANSWER 4:

Entities

Patient

- Patient_ID (PK)
- Name
- Age
- Gender
- Phone

Doctor

- Doctor_ID (PK)
- Name
- Specialization

Department

- Department_ID (PK)
- Department_Name

Appointment

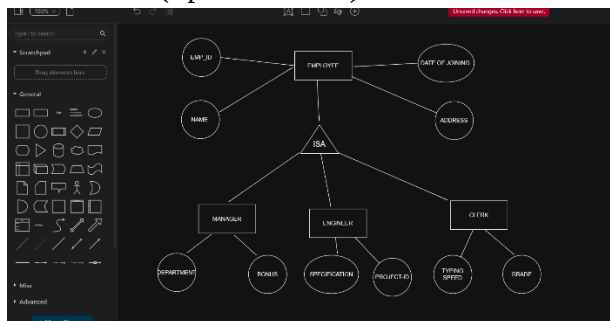
- Appointment_ID (PK)
- Date
- Time

Bill (Weak Entity)

- Bill_No
- Amount
- Payment_Status

Bill depends on Patient.

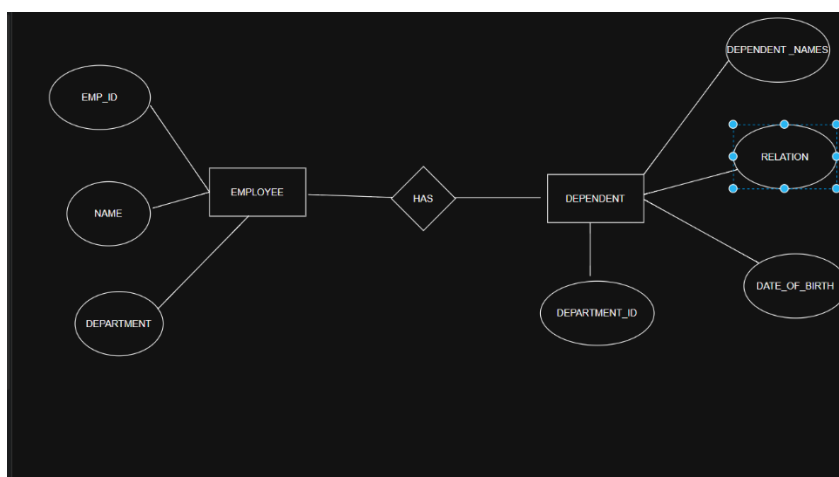
Inheritance (Specialization)



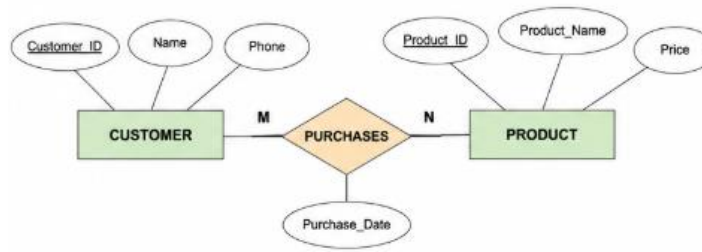
Person is the superclass.

Student and employee are subclasses.

Weak Entity



Simple EER Diagram



Explanation

- **Person** is the superclass.
- **Doctor** and **Patient** inherit common details.
- **Bill** is a weak entity because it depends on Patient.
- Doctor treats Patient through Appointment.

ANSWER 5:

Definition

A **Database Administrator (DBA)** is a person who manages, maintains, secures, and controls the database to ensure it works efficiently.

Responsibilities of DBA

1. Database Installation

- Installs and configures database software.

2. Database Design

- Creates tables, relationships, and schemas.

3. Security Management

- Creates user accounts.
- Grants and revokes permissions.

4. Backup and Recovery

- Takes regular backups.
- Restores lost data when required.

5. Performance Tuning

- Improves database speed using indexes and optimization.

6. Data Integrity

- Ensures accurate and consistent data.

7. Database Monitoring

- Monitors storage, memory, and performance.

8. Maintenance

- Updates database software and fixes issues.

9. Disaster Recovery

- Recovers the database after system failure.

10. User Support

- Helps users solve database-related problems.

Advantages of a DBA

- Ensures database security.
- Prevents data loss.
- Improves database performance.
- Maintains data consistency.
- Supports multiple users efficiently.